

INF115 Databases and Modelling

Obligatory Assignment 1

Deadline: Monday 9th of March, (18:00)

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Submission

This assignment deals with the use of entity relationship (ER) diagrams and normalisation in database design. You must submit your solution as a single zip-file containing exactly two files:

- oblib1.pdf containing solutions to Problems 1–4 and
- bysykke1.db,

before the deadline.

We recommend using Lucidchart (<https://www.lucidchart.com/>) or Draw.io (<https://www.drawio.com/>), to make the ER diagrams. You can download individual diagrams as png images that can then be combined into a single pdf document. Primary keys should be marked with “(PK)” (see Fig. 1). If you are using Lucidchart you should enable the UML diagram tool (https://www.lucidchart.com/pages/examples/uml_diagram_tool).

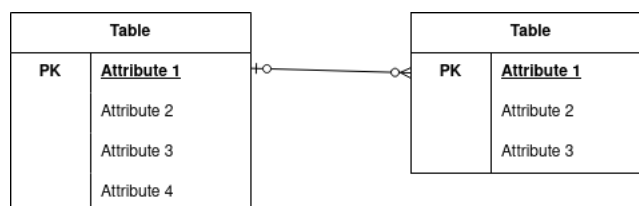


Figure 1: An example ER diagram.

For some problems you will be asked to create tables. We give an example below to show the format you should use. Mark primary keys with “#” and foreign keys with “*”.

- Table_1(#Attribute_1, Attribute_2, Attribute_3)
- Table_2(#Attribute_1, *Attribute_3)

Important! If you work together in groups, **everyone** in the group must submit the assignment on mitt.uib. In addition you must clearly state the names of all the colleagues that you worked with on the first page of the document. You may use various sources to help you solve the problems, but remember to cite the sources in the code (as comments) when significant parts of the code are taken from sources. Deliver a clearly written document, with headings for each question answered, example:

3.1

answer or figure...

3.2

answer or figure...

Points will be deducted if formatting is unclear. Late submissions will be penalized with a 10% point deduction for each late day.

Good luck!

Tips

Are you struggling with the tasks?

- Stay Calm!
- Recheck your notes and the slides.
- Go to your group sessions and ask your questions there.
- Ask on Discord server.
- Search the internet as all programmers do.

Introduction

You have been hired to create a database design for the leading bike commuting company in Bergen, Bergen Bysykkel. They need you to provide specific ER diagrams to developers such that they can implement your design. They give you the following information on the elements of the Bysykkel operation. You must assess how each entity of the operation is to be represented in database design (ER-diagram). The following entities are given with their respective properties:



Bergen Bysykkel

Bike

- Each bike must have a unique ID-number
- The bikes name, a random name which are not necessarily unique (it could be Arne or Fatima)
- The last station it was registered at. If the bike is active, it is the station it left from.
- Current status. What status should the bike have? For example, "Active", "Parked" or "Missing"
- Reparation status (Users can send in complaints about the bikes, hence the reparation status can either be NULL or any number of complaints from a defined list of 12 possible complaints: ("flat tire", "breaks not working", "gear not working", ...). So this element can contain multiple values ("flat tire", "flat tire", "flat tire") given from 3 different users). When a mechanic has fixed the complaints given, the element is set back to NULL.

User

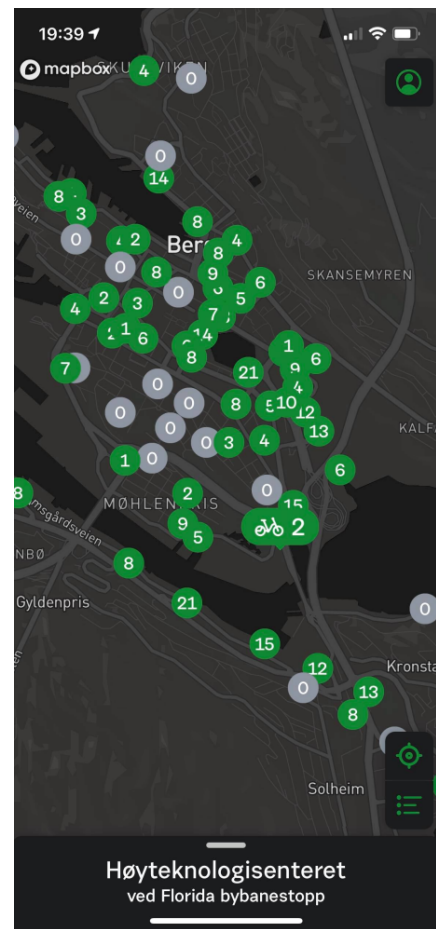
- A unique ID-number
- A name
- Their telephone number (Can several users have the same phone number?)
- Their GPS position (This is used to see where the nearest bicycle station is and is stored in latitude and longitude. For example in SQL:
 - Latitude `FLOAT(10,6)`
 - Longitude `FLOAT(10,6)`

Station

- A unique ID-number
- A station name
- Its GPS coordinates
- Maximum parking spots
- Available parking spots

Problem 1

1. Create an ER diagram from the description given above. The diagram should represent the entities and their attributes. Mark primary and foreign keys, and include relational arrows.
2. Given the current three entities, do all three tables participate in a relationship. Example: does a user own a bike?

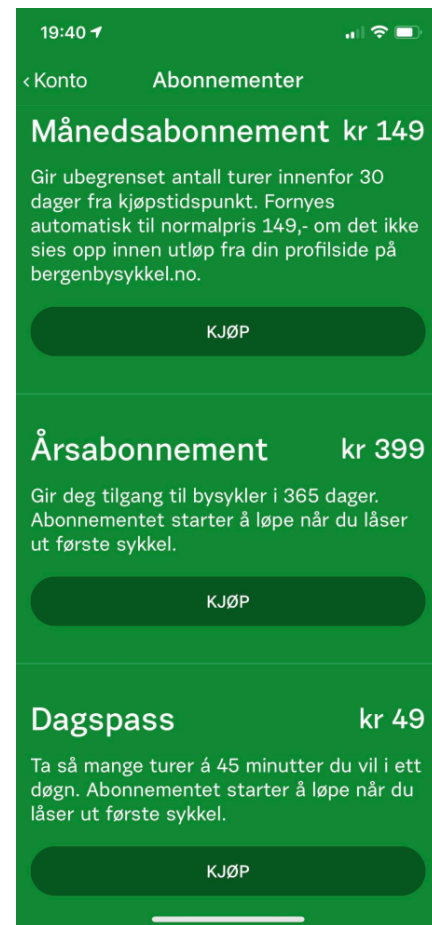


The Bysykkkel app

Problem 2

Bergen Bysykkkel have now figured out that they want further improvements to their database design. Until now the design has not included handling of subscriptions. This must be implemented to keep track of who has currently got a membership and who have had it before. A user can have more than one membership (the 1 active and older inactive ones). A fellow junior developer proposes that we add the following attributes to the User entity:

- SubscriptionID
 - Status
 - Start
 - End
 - Type
1. What is the problem with the proposed solution? Are there unwanted dependencies if we add the attributes that the junior developer proposes? Identify the problem.
 2. Create table(s) to solve the problem. Mark primary keys, and foreign keys.
 3. Based on the tables created, produce the extended ER diagram. Include appropriate primary, foreign keys and relations.

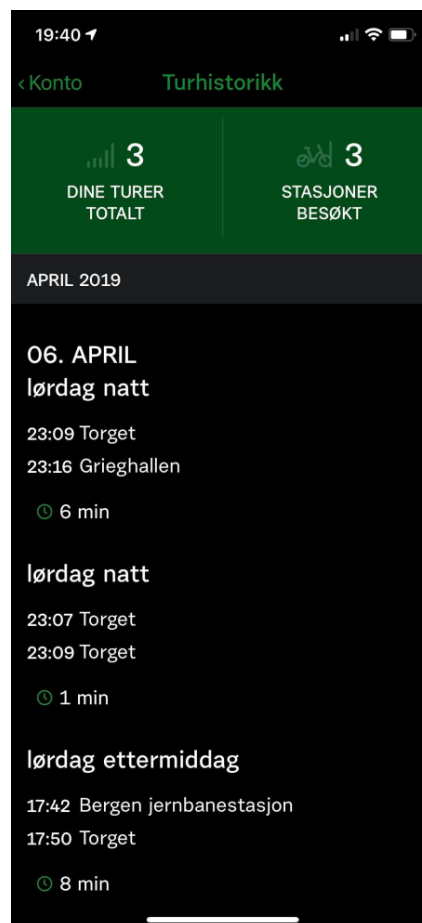


Subscription options

Problem 3

Great! Directors at Bergen Bysykkel like your solution to the Subscription problem, and have decided that you are promoted to Lead Developer, and they have given you the responsibility of the next big problem to tackle with building their app. The design, in its current state, does not register who uses each bike. This is a problem and you now have to design the **Trip** entity to take this into account. A Trip has to keep track of the user, the bike, the start time, the end time, the start station and the end station.

- (a) Identify the attributes in the database description above and create a table.
- (b) Extend your ER diagram with this new entity and represent the relationships between the entities that you identified in the previous question. Specify the primary keys and choose appropriate relationships between the entities.
- (c) The current database design is taking too much storage (some of your tables contain redundant information). Luckily you learned in INF115 lectures that there are 4 normal forms to reduce size of a database system. First, explain the four normal forms (1NF, 2NF, 3NF, BCNF).
- (d) Your boss wants you to normalise the database to the BCNF. Describe what changes you would have made to the current design to achieve this, and reduce redundancy. Then create tables to your suggestion. (You don't need to extend your ER-diagram) Mark primary keys and foreign keys. Explain the changes you make. Example: Is Reparation Status atomic (1NF)?



Trips

Problem 4

Now that the design phase is done, it is time to implement the app. You are asked to build a database.

1. Create a `bysykke.db` database based on your answers to Problems 1–3.
2. Using the data from `bysykke1.csv` populate the tables in your database.
3. How many times was each subscription type purchased? Answer with both a SQL query and its result.